

Killian Steunou

Computer Vision Researcher / Efficient Video Understanding

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Summary

PhD student with **Institut Polytechnique de Paris** (SAMOVAR, Télécom SudParis) and Research Scientist at **Moments Lab**, working on efficient video understanding. Recent work on frame selection for video-language models (**PEEK**) was accepted at **BMVC 2026**.

Education

- 2025–2028 **PhD in Machine Learning**, *Institut Polytechnique de Paris*, Palaiseau, France.
SAMOVAR, Télécom SudParis — *Efficient Omni-Modal Learning for Generalized Video Understanding*.
Advisors: Mounîm A. El-Yacoubi and Yannis Tevissen.
- 2024–2025 **M2 Mathématiques, Vision, Apprentissage (MVA)**, *ENS Paris-Saclay*, Gif-sur-Yvette, France.
Optimal Transport, Computer Vision, Probabilistic Graphical Models, Representation Learning, Generative Models, 3D Modeling
- 2022–2025 **Magistère d'Économiste Statisticien**, *Toulouse School of Economics*, Toulouse, France.
Highest honors, ranked 1/16
- 2023–2024 **M1 Applied Mathematics & Statistics**, *Toulouse School of Economics*, Toulouse, France.
Econometrics, Probability, Optimization, Machine Learning, Game Theory, MDP
- 2019–2022 **Double Bachelor, Applied Mathematics & Economics**, *Toulouse School of Economics*, Toulouse, France.
Gap semester at the University of Copenhagen, Denmark (2022).

Experience

- Nov 2025–Present **Research Scientist | PhD**, *Moments Lab*, Paris, France.
Research on efficient omni-modal learning for video understanding: frame selection, multimodal representation learning, and evaluation of video-language models. First-author publication at BMVC 2026.
- Apr 2025–Oct 2025 **Deep-Learning Research Intern**, *IDEMIA*, Paris, France.
 - Research on multi-object tracking and segmentation.
 - Exploring various ways to use the SAM 2 foundation model for training end-to-end MOT models
 - Create video instance segmentation data from MOT data
 - Train an end-to-end tracker with an instance segmentation branch on such data
 - Use SAM 2 embeddings as instance representations during training
- Apr 2024–Aug 2024 **AI Research Intern**, *C.L.S.*, Toulouse, France.
 - Benchmarked foundation models for Earth observation, on the task of semantic segmentation.
 - Surveyed state-of-the-art self-supervised methods for imagery and remote sensing.
 - Developed a Python library for fine-tuning vision foundation models, that led to a poster presentation.
- Feb 2023–Jul 2023 **Machine-Learning Engineer Intern**, *Jolibrain*, Toulouse, France.
 - Contributed to the open-source image-generation tool joliGEN.
 - Implemented edge-detection methods for image generation control.
 - Trained experimental diffusion models for inpainting.

Publications

- [1] **K. Steunou**, Y. Tevissen, and M. A. El-Yacoubi, Why Is Video Still So Expensive? A Survey of Inference-Efficiency Mechanisms in Video and Audiovisual LLMs, in *arXiv preprint arXiv:2609.10355*, Sept. 2026.
- [2] B. Roussel, **K. Steunou**, D. Martin, and P. Perrault, Multi-Object Tracking as Identity Correction, *Preprint*, July 2026. [Article](#) .

- [3] **K. Steunou**, A. Filali Razzouki, K. Guetari, M. A. El-Yacoubi, and Y. Tevissen, PEEK: Picking Essential frames via Efficient Knowledge distillation, in *BMVC*, June 2026. [Article](#) - [Github Repo](#).
- [4] **K. Steunou**, T. Druilhe, and S. Saue, Sparse Representations Improve Adversarial Robustness of Neural Network Classifiers, *arXiv preprint*, 2025. [Article](#).

Open-Source & Releases

PEEK	Frame-selection model, training code and weights. Code Weights Demo
joliGEN	Contributor to Jolibrain's open-source image-generation toolkit. Github
Video BG Removal	Automatic background removal behind subjects in video. Github

Selected Projects

- Jan 2025 **Score-based Generative Networks for Large-Scale Optimal Transport.**
Extended the SCONES method: validated it on low-dimensional data and characterized the impact of several hyperparameters. [Github](#) | [Report](#)
- Dec 2024 **Test-Time Training with Masked Autoencoders.**
Online extension of TTT-MAE that outperforms the original method. [Github](#) | [Report](#)

Skills

Languages	Python, R, Bash, $\text{\LaTeX}$
Deep learning	PyTorch, PyTorch-Lightning, Hugging Face, Accelerate
Tooling	Git, Docker, ffmpeg, NumPy, Pandas, Streamlit, multi-GPU
Spoken	French (native), English (professional)